

# Zacharia Hammad

Software Engineer | AI Products & Developer Tools

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Software engineer building AI products and developer tools in Python, TypeScript, and Rust, with experience in computer vision, voice agents, and graph-backed applications.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, Rust, SQL, Tcl

**Frameworks and APIs:** React, Next.js, FastAPI, LLM APIs, VAPI, OpenAI Realtime

**Data and ML:** PostgreSQL, Neo4j, Kafka, YOLO, ONNX, NVIDIA DeepStream, MLflow

## PROFESSIONAL EXPERIENCE

**Visionary Solutions** | Aug 2025 - Present

Software Engineer

- Owned computer-vision product development across data sourcing, labeling, model training, and the user-facing application.
- Implemented shared TypeScript tool execution for VAPI and OpenAI Realtime voice agents, tying actions to verified caller identities.
- Built Neo4j identity matching and conversation context with access controls for AI applications.
- Developed YOLO/ONNX training and release tooling with MLflow tracking, evaluation, and model promotion and rollback; integrated NVIDIA DeepStream inference.
- Implemented event delivery between services using PostgreSQL and Kafka.

**PECO / Exelon** | Sep 2023 - Mar 2024

UAS Engineering Co-op

- Automated drone reports with Python and APIs, reducing preparation time from two hours to under 30 minutes.

**NAVSEA** | Sep 2022 - Mar 2023

Engineering Co-op

- Improved submarine simulation algorithms and sensor models; documented gas-sensor workflows.

**Saint-Gobain** | Sep 2021 - Mar 2022

Research and Development Co-op

- Analyzed materials experiments with Python, MATLAB, and statistics to identify process improvements.

## PROJECTS

**RepoToire** | Rust and Quint

- Developed a Rust runtime prototype for AI coding agents with validated execution plans, staged execution, bounded retries, and Quint-to-Rust conformance tests.

**ML Pothole Detection System** | Drexel Senior Design

- Developed YOLO edge inference with GPS-tagged capture, backend APIs, PostgreSQL, and a geospatial dashboard for a Drexel Senior Design Championship-winning team project.

**ASIC Design Optimization** | Drexel Course Project

- Adapted Tcl scripts for Synopsys design flows in a two-person course project, comparing timing and area across five benchmark circuits using SkyWater and ASAP7 libraries.

## EDUCATION

**Drexel University** | Jun 2025

B.S. Computer Engineering | Minor in Data Science | GPA 3.69

## CERTIFICATIONS

**Neo4j Certified Professional** | Aug 2026